

2(a)(i)	the higher the DDD (of antibiotic / penicillin / macrolide) the higher the <u>percentage</u> of resistant bacteria / AW ; lower DDD macrolide give greater percentage of resistant bacteria (than penicillin) ;	2
2(a)(ii)	11:3 / 3.7 :1 / 3.67:1 / 3.6 recurring : 1 ;	1
2(b)(i)	<i>any two from</i> 1 (both sets of) data are quantitative ; 2 more than 5 paired observations / AW ; 3 data shows a normal distribution ; 4 if data were plotted (on a scatter diagram) would show a linear relationship ;	2
2(b)(ii)	<i>any two from</i> 1 there is <u>positive correlation</u> (between quantity of antibiotics used and percentage resistance to the antibiotic) ; 2 penicillin has a stronger correlation than macrolides ; ora 3 less than 1% probability / chance that the correlation is due to chance ; or more than 99% certain / confident / sure that the resistance is due to the antibiotic used / DDD ;	2
2(c)	put a standard volume of stock solution into a standard volume of water ; use new dilution for making the next dilution in the same way as above / AW ;	2

Question	Answer	Marks
2(d)	<i>any two from</i> 1 reliable / reproducible / AW ; 2 <i>idea of.</i> sensitive / able to detect low numbers (of resistant <i>E. coli</i>) ; 3 specific (to resistant <i>E. coli</i>) / able to detect resistant <i>E. coli</i> ; 4 cost effectiveness ; 5 <i>idea that:</i> results are similar to the disc diffusion results ;	2